Project: Statistical Methods for Climate Science and Machine learning (MO7019)

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Objective

The objective is to analyse the modes of variability of sea level pressure from NOAA. The (netCDF file) is mslp.mon.mean.nc, which can be downloaded from
http://www.esrl.noaa.gov/psd/data/gridded/data.ncep.reanalysis2.surface.html

Task I

Read the data into Python (or the software of your choise, e.g. Matlab).
Calulate the annual means of the SLP field at each grid point.
Calculate the climatologies for the first and second halves of the record (at each grid point).
Analyse the difference between these climatologies and its significance (can take $\alpha = 5\%$).
Plot a map of the difference (second half minus first half) and the result of the significance.

Task II

Calculate the seasonal cycle (at each grid point) and obtain the SLP anomalies by subtracting from each month its climatology.
Calculate the EOFs of the winter (DJF) means of SLP anomalies over the Northern Hemisphere north of 20^oN.
Discuss the leading EOF and the corresponding PC (EOF1/PC1), and analyse the spectrum of PC1

Task III

Next consider the tropical belt between say -20 and +20. Take the average of monthly anomalies over two region centered respectively around Tahiti (17.65S, 149.4W) and Darwin (12.46S, 130.84E) – Can take closest 9 grid points to each - Note that Darwin and Tahiti will most probably not be on the grid points of the data).

Analyse the power spectrum of the obtained time series and discuss if you believe there is some power at specific frequecies (or periods).

Claculate the correlation between the obtained Darwin (resp. Tahiti) time series and every grid points. Plot the correlation maps, and discuss the obtained results.

Calculate the regression between the obtained Darwin time series and the time series of SLP anomalies at each grid point. Plot the obtained map with significance, and discuss the obtained result. In this cas, you can consider Darwin as predictor in the first case and Darwin as predictand in the other case, and discuss whether there is a difference between the two cases (and if there is a difference.)

Next choose either task VIa or VIb

Task VIa

Back to the SLP anomalies in in the northern hemisphere, north of 20N. Apply the k-means clustering with $k = 2$ and get the cluster centers. Apply hierarchical clustering and plot the dendrogram, and obtain two clusters. Compare the clusters obtained by both methods.

Task VIb

Consider again the northern hemisphere SLP , north of 20N. Apply self organizing map (SOM) to the SLP anomalies, and show the array of the obtained organised maps. Discuss whether the data can be clustered into a number of clusters.

Task V - for PhD students

Consider the low-order chaotic Lorenz' system:

$$\begin{aligned}
\dot{x} &= \tfrac{dx}{dt} = \sigma(x - y) \\
\dot{y} &= \tfrac{dy}{dt} = rx - y - xz \\
\dot{z} &= \tfrac{dz}{dt} = xy - bz
\end{aligned}$$

with parameters $\sigma = 10, r = 28, b = 8/3$. The objective is to compare few forecasting methods.

a. Integrate the model forward in time with a time step $\Delta t = 0.1$ for a time interval $T = 10^4 \Delta t$. Plot the time series of x and the phase plot in (x, z) plane.

b. Subsample the time series with a time interval $\tau = 5\Delta t$. Then consider the following forecasting methods:

Random forest

Neural network (NN)

LSTM

Gaussian process

Split the time series into series, a training set and a testing set. The training set is used to fit the regression model and the testing set is used to test the model. Use correlation analysis and RMSE measures of performance between forecasts and observations obtained from the testing set for various lead times.